Homework 4

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Instructions

Use functions learned from the tidyverse package to complete this homework assignment. Load all the packages you will use throughout this homework in the loading packages section below.

Submit the html file or PDF in Sakai before the deadline.

AI Declaration

(15 points)

You have to declare whether you did or did not use AI. If you use AI, you have to specify following things:

1. For which questions you used AI?

Answer part a-e for every question in which you used AI

a. AI Tool Used: [name and version]
b. Purpose: [brief description]
c. Learning rationale: [why this supported your learning]
d. Questions you asked AI
e. Consolidate you AI questions and output you got from AI in a single file and submit as an attachment. (mention here if you submit such pdf)

Please refer to AI policy before you use any AI.

If your code is a subset of your AI output, you will receive zero on such assignment.

If you use AI and do not submit your AI history, mentioned in part e, you will receive 50% credit.

Add your AI declarations here and submit part e as a separate file on sakai:

AI Declarations:

I DID NOT USE AI AT ALL

Loading Packages

(5 points)

```
library(tidyverse)
library(mdsr)
library(Lahman)
```

Exercise 1

Use the `Batting`, `Pitching`, and `People` dataframes in the Lahman package to answer the following questions. Make sure that you read the background information on these dataframes and understand them well before answering each question.

a. (15 points) Name every player (first and last name) in baseball history who has accumulated across their careers 300 home runs (HR) or more AND 300 stolen bases (SB) or more. You can find the first and last name of the player in the `People` data frame. Join this to your result along with the total home runs and total bases stolen for each of these elite players. ***Your final output should be a printed dataframe that has four columns reflecting the players' first name and last name and the total homeruns and total stolen bases***

```
data(People)
data(Batting)

databs<- Batting %>%
  select("HR", "playerID", "SB")

dataps<- People %>%
  select("playerID", "nameFirst", "nameLast")


combined_data<- databs %>%
 left_join(dataps, by=c("playerID" = "playerID"))


questiona<- combined_data %>%
  group_by(playerID, nameFirst, nameLast) %>%
  summarise(total_HR = sum(HR), total_SB = sum(SB)) %>%
  filter (total_HR >= 300)%>%
  filter (total_SB >= 300)

questiona
```

```
## # A tibble: 8 × 5
## # Groups:   playerID, nameFirst [8]
##   playerID  nameFirst nameLast  total_HR total_SB
##   <chr>     <chr>     <chr>        <int>    <int>
## 1 beltrca01 Carlos    Beltran        435      312
## 2 bondsba01 Barry     Bonds          762      514
## 3 bondsbo01 Bobby     Bonds          332      461
## 4 dawsoan01 Andre     Dawson         438      314
## 5 finlest01 Steve     Finley         304      320
## 6 mayswi01  Willie    Mays           660      339
## 7 rodrial01 Alex      Rodriguez      696      329
## 8 sandere02 Reggie    Sanders        305      304
```

b. (15 points) Similarly, name every pitcher (first and last name) in baseball history who has accumulated 300 wins (W) or more AND 4,000 strikeouts (SO) or more. ***Your final output should be a printed dataframe***

that has four columns reflecting the players' first name and last name and the total wins and total strikeouts

```
data(Pitching)

questionb<- Pitching%>%
  group_by(playerID)%>%
  summarise(total_W = sum(W), total_SO = sum(SO))%>%
  filter (total_W >= 300)%>%
  filter (total_SO >= 4000)

totalanswer<- questionb %>%
  left_join(People, c("playerID" = "playerID"))%>%
  select (nameFirst, nameLast, total_W, total_SO)

totalanswer
```

```
## # A tibble: 4 × 4
##   nameFirst nameLast total_W total_SO
##   <chr>     <chr>      <int>    <int>
## 1 Steve     Carlton      329     4136
## 2 Roger     Clemens      354     4672
## 3 Randy     Johnson      303     4875
## 4 Nolan     Ryan         324     5714
```

c. (15 points) Identify the name of the players who have hit 50 home runs or more in a single season. Of those players, which player had the lowest batting average (hits/at bats [H/AB])? ***Your final output should be a printed dataframe that has five columns and one row reflecting the one player's first name, last name, the year of the season, the number of homeruns, and the batting average (H/AB)***

```
smallbatting<- Batting %>%
  select(playerID, H, AB, HR, yearID) %>%
  group_by (playerID, yearID)%>%
  summarise (sumH = sum(H), sumAB = sum(AB), sumHR = sum(HR))%>%
  filter(sumHR >= 50)%>%
  mutate(battingavg = (sumH/sumAB))%>%
  arrange(battingavg)

questionc<- smallbatting%>%
  left_join(People, c("playerID" = "playerID"))%>%
  select(nameFirst, nameLast, battingavg, playerID, sumHR)


questionc
```

```
## # A tibble: 54 × 5
## # Groups:   playerID [34]
##    nameFirst nameLast  battingavg playerID  sumHR
##    <chr>     <chr>          <dbl> <chr>     <int>
##  1 Kyle      Schwarber      0.240 schwaky01    56
##  2 Cal       Raleigh        0.247 raleica01    60
##  3 Pete      Alonso         0.260 alonspe01    53
##  4 Jose      Bautista       0.260 bautijo02    54
##  5 Andruw    Jones          0.263 jonesan01    51
##  6 Roger     Maris          0.269 marisro01    61
##  7 Greg      Vaughn         0.272 vaughgr01    50
##  8 Mark      McGwire        0.274 mcgwima01    58
##  9 Cecil     Fielder        0.277 fieldce01    51
## 10 Mark      McGwire        0.278 mcgwima01    65
## # ℹ 44 more rows
```

Exercise 2

Use the `nycflights13` package and the `flights` and `planes` tables to answer the following questions. Make sure that you read the background information on these dataframes and understand them well before answering each question.

a. (15 points) What is the oldest plane (specified by the tailnum variable) that flew from New York City airports in 2013 and how many flights did this plane conduct from New York City airports in 2013? ***Your final output should be a printed dataframe that has three columns reflecting the tailnumber of the plane, the year it was manufactured, and the number of flights this plane conducted from NYC***

```
library(nycflights13)
library(tidyverse)
data(planes)
data(flights)

smallflights<- flights %>%
  select(tailnum, year, origin)%>%
  arrange(tailnum)

smallplanes<-planes%>%
  arrange(year)%>%
  rename (yearmade = year)

combineddata<- smallplanes %>%
  full_join(smallflights, by=c("tailnum" = "tailnum"))

question2a<- combineddata%>%
  select(tailnum, yearmade)%>%
  group_by(tailnum, yearmade) %>%
  summarize(total_flew=n())%>%
  arrange(yearmade)

question2a
```

```
## # A tibble: 4,044 × 3
## # Groups:   tailnum [4,044]
##    tailnum yearmade total_flew
##    <chr>      <int>      <int>
##  1 N381AA      1956         22
##  2 N201AA      1959         54
##  3 N567AA      1959         63
##  4 N378AA      1963         13
##  5 N575AA      1963         39
##  6 N14629      1965          4
##  7 N615AA      1967         22
##  8 N425AA      1968         43
##  9 N383AA      1972         25
## 10 N364AA      1973         22
## # ℹ 4,034 more rows
```

the plane with the tailnum N381AA is the oldest plane and flew out of NYC airports 22 times in 2013

b. (10 points) How many ***different*** airplanes (specified by the tailnum) are in the `flights` dataset? How many ***different*** airplanes (specified by the tailnum) are in the `planes` dataset?

YOUR WRITTEN ANSWER HERE

```
questionflights<- flights%>%
  group_by(tailnum)%>%
  summarize(diffplanes=n())

questionplanes<- planes%>%
  group_by(tailnum)%>%
  summarize(diffplanes2=n())
```

There are 4044 different planes in the flights dataset There are 3322 different planes in the planes dataset

c. (10 points) In 2013, there were 336,776 flights that left New York City, how many of those airplanes are included in the planes table? What does this imply if we do an inner_join with the `flights` and `planes` dataset by `tailnum` ? (You don't need to do an inner join, simply explain the implication).

YOUR WRITTEN ANSWER HERE There is a total of 284, 170 flights that left NYC that are in both two datasets. But there is also 52606 flights that are in either one of the other, so this means that innerjoin is only helpful when there is the same amount of certain rows.

```
table(flights$tailnum%in%planes$tailnum)
```

```
##
##  FALSE   TRUE
##  52606 284170
```

```
bothsets<- planes%>%
  inner_join(flights, by=c("tailnum"="tailnum"))
```